

BUILD YOUR PERSONAL CYBERSECURITY LAB

Task-2

Objective

The objective of this task is to help you **build a safe and isolated cybersecurity practice environment** on your own computer. This hands-on lab will serve as the foundation for all future tasks such as vulnerability assessment, web application testing, exploitation practice, and incident response.

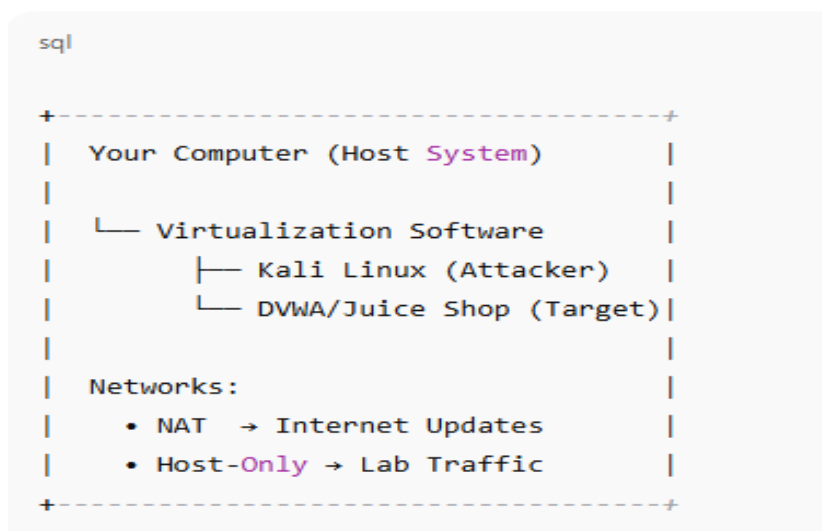
You will create:

- 1 Attacker machine (Kali Linux)
- 1 Vulnerable target application/machine (DVWA or OWASP Juice Shop)
- Proper network segmentation (NAT + Host-Only)

What You Will Learn

Skill Area	Practical Outcome
Virtualization	Running multiple isolated systems for pentesting
Lab isolation	Preventing real-world system damage
Web targets setup	Deploying intentionally vulnerable applications
Basic networking	NAT vs Host-Only and subnet usage
Tool awareness	Nmap, Burp, Wireshark validation

Architecture (Visual Overview):



Prerequisites

Requirement	Recommendation
RAM	Minimum 8GB, Preferably 16GB
Disk Space	40–60 GB free
CPU	Virtualization enabled
OS	Windows / Mac / Linux

Software (Official Links)

Tool	Official URL
VirtualBox	https://www.virtualbox.org/wiki/Downloads
VMware Workstation Player (optional)	https://www.vmware.com/products/workstation-player.html
Kali Linux VM	https://www.kali.org/get-kali/#kali-virtual-machines
OWASP Juice Shop	https://github.com/juice-shop/juice-shop
DVWA (Damn Vulnerable Web App)	https://github.com/digininja/DVWA
Docker (for Juice Shop deployment)	https://docs.docker.com/engine/install/

Step-by-Step Instructions

✓ Step 1: Install VirtualBox / VMware

Install VirtualBox → Create Host-Only Network → Enable NAT.

✓ Step 2: Deploy Kali Linux (Attacker VM)

- Import prebuilt VM image
- Assign 2 CPUs, 3–4GB RAM
- Enable two network adapters:
 - Adapter 1 → NAT
 - Adapter 2 → Host-Only
- Update system:

```
sql

sudo apt update && sudo apt upgrade -y
```

- Create working folders:

```
bash

mkdir -p ~/lab/{scans,notes,pcaps,screenshots}
```

✓ Step 3: Deploy the Target (Choose One)

Option A: OWASP Juice Shop via Docker

```
bash

sudo apt install -y docker.io
sudo systemctl enable --now docker
sudo docker pull bkimminich/juice-shop
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

Option B: DVWA

Setup Apache, PHP, MariaDB + clone repo. (Instructions available in official DVWA GitHub)

✓ Step 4: Verify Connectivity

```
nginx

nmap -sn 192.168.56.0/24
```

Confirm attacker machine sees target.

✓ Step 5: Basic Validation

- Access target web app in browser
- Open Burp and intercept traffic
- Run a simple Nmap scan

Free Learning Resources (Beginner-Friendly)

Topic	Link
VirtualBox Setup	https://youtu.be/w-1B_Y6KJ-U
Install Kali Linux	https://youtu.be/GeyZ4A1TgOE
What is OWASP Juice Shop	https://youtu.be/2cB5XNhAovA
DVWA Setup Tutorial	https://youtu.be/NPqerU5WqfM
Burp Suite Basics	https://portswigger.net/burp/documentation/desktop
Wireshark Basics	https://www.wireshark.org/docs/

Deliverables

- Full Lab Build PDF Report (including screenshots)
- Evidence folder containing:
 - VM list
 - IP configuration
 - Ping/Nmap validation
 - Burp + Wireshark screenshots